

The Alternative Hypothesis, Finitely:

What the New Finite Theorems Say About Tao's ACUE Problem
and the Unconditional Two-Thirds Theorem

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Abstract. These notes are written for a working mathematician who knows what the Riemann Hypothesis is but has not followed the Alternative Hypothesis (AH) literature. We explain, in plain terms: (i) Tao's 2019 construction of the Alternative Circular Unitary Ensemble (ACUE) and the inverse problem it poses; (ii) a body of recent finite theorems about that inverse problem — finite non-uniqueness of the ACUE moment problem, a rigidity ladder (two- and three-point correlations frozen, freedom starting at four points), an exact “Nyquist conservation law” locating all multiplicity in a single Fourier mode, and a proved transfer principle from finite mimickers to the Lagarias–Rodgers uniqueness question; and (iii) why these finite facts are the correct lens for understanding what the 2026 unconditional theorem — at least 67.25% of zeta zeros are simple and on the critical line — did, and why it stopped where it stopped. All specific finite claims cited here have been independently re-verified (by hand proof, exact enumeration, or high-precision computation) in the companion document *Finite Spectral Certificates and Their Fibers*. Nothing here is a theorem about zeta zeros; we are explicit about significance levels throughout.

1. Tao's problem: the enemy has a name and a matrix model

The Montgomery–Odlyzko phenomenon says that the normalized gaps between zeros of the Riemann zeta function look statistically like the gaps between eigenvalues of a large random unitary matrix (the CUE, equivalently GUE locally). Montgomery's 1973 theorem is the rigorous core: under RH, the pair correlation of the zeros, tested against functions whose Fourier transform is supported in $(-1, 1)$, matches the random-matrix prediction. Everything sharper is conjecture.

The **Alternative Hypothesis** is the nightmare scenario compatible with that theorem: the normalized gaps between consecutive zeros cluster near half-integers $0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$ — a rigid, lattice-like, profoundly non-random spacing law. It is not a fringe worry. If Landau–Siegel zeros exist (infinitely many, in a suitable sense), work in the tradition of Conrey–Iwaniec shows the zeta zeros would exhibit exactly this kind of alternative spacing. So “rule out AH” is at least as hard as the Siegel-zero problem — one of the hardest problems in analytic number theory. AH is the precise reason pair-correlation methods cannot be pushed to 100% of anything: every statistic those methods can see, AH can imitate.

In his May 2019 blog post “The alternative hypothesis for unitary matrices”, Tao gave the enemy a concrete random-matrix body. Fix N and place $2N$ equally spaced sites on the unit circle (the $2N$ -th roots of unity, times one uniform random global rotation).

Choose N of the $2N$ sites with probability proportional to the squared Vandermonde determinant — the natural discrete log-gas at inverse temperature $\beta = 2$. Call the resulting N -point process the **ACUE**. Equivalently, the ACUE is the determinantal process whose kernel is the rank- N Fourier projection onto N consecutive frequencies — a perfect finite “Fermi sea.” Its gaps are, by construction, half-integer multiples of the mean spacing: this is AH, sampled from a matrix model.

Tao's matching theorem. The ACUE is indistinguishable from the CUE (Haar-random $U(N)$) across every balanced trace moment of weighted degree at most N : for partitions λ, ν with $|\lambda| = |\nu| \leq N$, $E[p_\lambda \bar{p}_\nu]$ takes the Diaconis-Shahshahani CUE values $\delta_{\lambda\nu} z_\lambda$ (where $p_k = \text{tr } U^k$). Consequently all band-limited linear and correlation statistics of bounded order agree. Lagarias-Rodgers (2020, 2021) proved the infinite-volume counterpart: an AH-type process on the half-integer lattice matches the sine process against all band-limited correlation tests of every order. Band-limited data — of any correlation order — cannot kill AH.

Tao's construction converts a vague fear into two sharp inverse problems, which the recent work attacks directly:

Q1 (identifiability). Exactly which statistics pin down the ACUE? Given all the moments it shares with CUE, is the ACUE the unique law with those moments — or is it one point of a large fiber?

Q2 (uniqueness at infinity — Lagarias-Rodgers). Is the AH point process the unique process matching the sine process in all band-limited correlations with half-lattice-supported gaps? If not, the AH scenario is even less determined than feared, and the enemy is a whole family.

2. What is now known about Q1: the moment data pin down almost nothing — except exactly what matters

2.1 Finite uniqueness is false, and massively so

Theorem (finite non-uniqueness; exact). For $N \leq 4$ the ACUE is the unique rotation-invariant law on N -subsets of $\mathbb{Z}/2N$ with its matching moments. From $N = 5$ on, uniqueness fails: there is an explicit counterexample law at $N = 5$ with coefficients in $\mathbb{Q}(\sqrt{5})$ (verified here by exact cyclotomic arithmetic: all 88 balanced moment pairs, positivity, everything), machine-certified counterexamples at $N = 9$ and $N = 10$, and the dimension of the fiber of “mimicker” laws grows super-exponentially:

$$\text{fiber dimension} = 0, 0, 2, 10, 80, 403, 1804 \text{ for } N = 3, 4, 5, 6, 7, 8, 9.$$

By $N = 9$ the mimickers fill two-thirds of the dimension of the whole orbit space. **Significance for Q1:** the answer is a resounding no — Tao's matching moments determine an exponentially thin slice of the law. The ACUE is a canonical point of the fiber (it is the unique determinantal one, see §2.3), but the moment data alone do not know that. Anyone who imagined “matching all CUE moments up to the natural bandwidth” as essentially characterizing the model must update.

2.2 The rigidity ladder: where the freedom starts

The fiber is enormous, yet strangely invisible. Verified exactly at $N = 5$ and to residuals $\sim 10^{-14}$ for $N \leq 9$: **every mimicker has identical two-point correlations at every distance** — not only the band-limited ones. The proof is a finite-group aliasing argument: on $\mathbb{Z}/2N$, every pair-count Fourier coefficient folds back ($k \mapsto 2N - k$) into the constrained band, so the “missing” high frequencies were never missing. **Three-point correlations are also frozen** across the entire fiber. The first genuine freedom appears at four-point statistics (at $N = 5$ the first distinguishable pattern is four consecutive empty sites).

Why a professor should care. This is the exact finite echo of Lagarias–Rodgers mimicry, with an address attached: any observable that distinguishes one mimicker from another — and hence any statistic that could ever separate an AH-like imposter from the canonical model — must be a correlation of order ≥ 4 , or must leave the band. Two- and three-point technology is structurally incapable of seeing the fiber. This localization was not in Tao's post and not in Lagarias–Rodgers; it is new, and it tells the field where not to dig.

2.3 The quasi-free gap: one row identifies the ACUE — inside the right class

There is a striking counterpoint, proved and verified. Restrict attention to stationary projection determinantal laws (finite Fermi seas: pick N Fourier modes S , form the projection kernel). Then the form factor is literally a boundary: $F(\gamma) = \frac{1}{2}|S \Delta (S+\gamma)|$, and $F(1)$ counts the occupied “runs” of S . Hence the single scalar $E|p_1|^2 = 1$ forces S to be one interval — i.e., **one number identifies the ACUE inside the determinantal class**. Outside that class, matching every inclusion correlation up to order $N - 1$ still leaves an invariant fiber of dimension $\sim 4^N/(2\sqrt{\pi} N^{5/2})$ (necklace-counting formula, verified). The determinantal (Wick) axiom is thus worth an exponential amount of information. Slogan: “the zeros have CUE pair statistics” and “the zeros behave like a determinantal process” differ by exactly this gap — an important corrective to how the GUE analogy is often used.

2.4 The Nyquist conservation law: where multiplicity hides

The deepest new finite theorem concerns marked configurations — nonnegative integer multiplicities m_x with $\sum m_x = N$ on $\mathbb{Z}/2N$, the right setting once one remembers that the real zeta question is about simple zeros. Impose the finite CUE ramp $E|p_k|^2 = k$ for $1 \leq k \leq N - 1$ and leave the edge (“Nyquist”) mode $k = N$ free. Three exact facts (each re-proved by hand and machine-verified here):

- (a) $E \sum m_x (m_x - 1) = (E|p_N|^2 - N)/2N$. All repeated-point mass is stored in the one unconstrained mode. Closing that single row forces every configuration to be simple.
- (b) The combination $s_1 - \sum m_x m_{x+1} - \sum h(m_x)$ (simple sites, minus nearest-neighbor overlap, minus a penalty for marks ≥ 3) is blind to the Nyquist mode — its Fourier

weight there is $1 + \cos \pi = 0$ — which yields the exact lower bound

$$E s_1/N \geq \frac{1}{2} + \csc^2(\pi/2N)/(2N^2) \longrightarrow \frac{1}{2} + 2/\pi^2 = 0.7026423673\dots,$$

with equality exactly on hard-core configurations with marks in $\{0, 1, 2\}$ and no two adjacent occupied sites — a face in explicit bijection with the binary configurations (it has exactly $C(2N, N)$ elements).

(c) At general density ρ the same mechanism gives the curve $\Phi(\rho) = 1 - \rho + \sin^2(\pi\rho)/(\pi^2\rho)$: the sine kernel's quadratic level repulsion becomes a cubic collision-rigidity loss at low density.

Why this matters for AH. The AH scenario's signature freedom — repeated zeros / multiplicity — occupies, in the finite model, exactly one Fourier coordinate, and that coordinate carries normalized weight $1/N$: it vanishes in every scaling limit. This is the precise finite mechanism behind the folklore that “the alternative hypothesis hides at the edge of the Montgomery band.” A statistic invisible in the limit rigidly controls the integer feasible set. In the conditional thermodynamic extension, the same scalar becomes a period-two Bragg atom whose mass is simultaneously a collision density and the squared norm of a $\mathbb{Z}/2$ dynamical eigenfunction — the AH's half-lattice rigidity, derived rather than postulated.

A caution the field should adopt. The constant $\frac{1}{2} + 2/\pi^2 \approx 0.70264$ looks temptingly close to the zeta constant 0.67250 below. They are not comparable: one is a lower certificate for simple sites in a marked finite moment model with an open edge row; the other is a proportion of simple zeta zeros proved from primes. Confusing the two feasible sets is the single most common error in informal discussions of this material (we made a version of it ourselves in an earlier synthesis, and retract it in the companion paper).

2.5 What can never work: three exact impossibility mechanisms

Three verified counterexample families draw the outer boundary of Q1. **Homometry:** two marked motifs on $\mathbb{Z}/12$ with identical complete squared-Fourier data (no extinctions) but simple-site fractions 0 and $\frac{1}{2}$ — so even complete pair/diffraction data cannot measure simplicity once multiplicity is allowed; the separation needs a cubic (bispectral, phase-sensitive) statistic. **Holonomy:** determinantal laws on a g -cycle agreeing in all correlations of order $< g$ whose total-variation distance is exactly $(2r)^g |\cos g\varphi - \cos g\psi|$ — bounded-order correlations miss a global phase entirely. **Local blindness:** a fiber of exponential dimension can project to zero on every fixed local cylinder statistic — so big fibers do not automatically produce observable gaps; a distinguishing observable must have support growing with N . Each of these is a regression test that any proposed “new statistic that kills AH” must pass.

3. Q2: a proved pathway to Lagarias-Rodgers non-uniqueness

The companion manuscript on finite non-uniqueness proves a clean **transfer theorem**: if there exist a fixed local pattern w , a constant $c > 0$, and finite mimickers q_N along $N \rightarrow \infty$ whose expectation of w differs from the ACUE's by at least c (a “uniform gap family”), then there exists a second infinite point process, distinct from the AH process, matching the sine process in all band-limited correlations of all orders — i.e., **the Lagarias-Rodgers uniqueness question would be answered in the negative**. No projective consistency across sizes is needed; a uniform finite gap suffices.

The quantitative state of play (verified): the maximal fiber gap $\delta^*(N)$ for the pattern 01010101 is increasing — 0.0102, 0.0123, 0.0138, 0.0151 at $N = 6, 7, 8, 9$ — and the hand-built $N = 9$ certificate achieves 97% of the LP optimum. The known obstruction (a strong-Rayleigh concentration bound forcing exponential density ratios for uniform-gap families) has no force until $N \approx 1.8 \times 10^7$; the certified counterexamples live comfortably below any such barrier. Against this: the rigidity ladder (§2.2) says any such gap is a ≥ 4 -point phenomenon, and spectral growth heuristics suggest $\delta^*(N)$ could rotate away from any fixed pattern. The question

$$\liminf_{N \rightarrow \infty} \delta^*(N) > 0 ?$$

is now a concrete, finite, machine-attackable optimization problem whose positive resolution settles a named open question of the field. In our assessment this — together with the all-size Nyquist equality problem (does the hard-core face support a law matching the full open ramp at every N ?) — is the most valuable single product of the entire program: Tao's and Lagarias-Rodgers' questions now have exact finite surrogates with explicit numerical frontiers.

4. The bridge to the two-thirds theorem

In August 2026, an Anthropic research model proved unconditionally, with a Lean-4 formalization, that at least

$$\delta_{MT} = 3/2 - (1/\sqrt{2}) \cdot \cot(1/\sqrt{2}) = 0.672500703679...$$

of the nontrivial zeros of ζ are simple zeros on the critical line (previous unconditional record: $5/12 \approx 41.7\%$). The decimal is the classical Montgomery-Taylor constant, known since the 1970s under RH. The advance is the removal of RH at zero cost: the proof measures finite compressions of Weil's explicit-formula quadratic form directly from the prime side (Montgomery-Vaughan quasi-orthogonality, unconditional pair-correlation input of Baluyot-Goldston-Suriyaya-Turnage-Butterbaugh, Bombieri's positivity analysis) and decodes positive inertia through rank-trace inequalities that exploit the integrality of zero multiplicities.

4.1 Why the AH is the reason the theorem says 67% and not 90%

The method's arithmetic input is confined to pair-correlation data of bandwidth one — precisely the data the ACUE/AH matches perfectly. The optimization over admissible windows is the Montgomery-Taylor variational problem (solved by the cosine profile $\cos(\sqrt{2} s)$; we verified the closed form and its finite-Galerkin ladder q_n with exact n^{-2}

error). An AH-type adversary is feasible for everything the certificate sees, and AH-type configurations carry multiplicity; so no certificate of this class can promise more than the AH-constrained extremum allows. In this exact sense, **Tao's ACUE is the reason the two-thirds theorem is a two-thirds theorem**: the constant is the price of fighting an enemy that band-limited pair data cannot distinguish from the truth. (The manuscript's own adversarial module reports that no bandwidth-one configurationwise certificate can exceed ≈ 0.68183 , so at most about one more percentage point is available to windowcraft — the rest requires new information. That ceiling currently rests on a hashed witness not yet replayed publicly; cite it as conditional.)

4.2 What the finite theorems add to the picture

First, an exact corollary worth knowing: since every distinct off-critical zero pair consumes two zeros while contributing no simple critical zeros, the theorem bounds the density of off-line “hyperbolic defects” of the Weil form by $(1 - \delta_{\text{MT}})/2 = 0.1637496\dots$. Second, the Nyquist mechanism (§2.4) is the finite-model explanation of the whole configuration: the multiplicity/AH freedom sits in a single edge mode that scaling limits erase, which is why (i) pair-correlation methods are blind to it, (ii) integrality — the one non-spectral input in the 2026 proof — is exactly what bites it, and (iii) improving on 0.6725 requires either Fourier support beyond bandwidth one (prime-pair information at twin-prime scale — and recall that killing AH outright is Siegel-hard) or genuinely new arithmetic positivity not reducible to correlations. The finite theory converts “why is this hard?” from folklore into checkable statements.

5. Honest significance assessment

For a skeptical colleague deciding whether any of this deserves attention, our grading — after independent verification of every checkable claim:

Result	Status	Significance for the AH problem
Finite non-uniqueness of the ACUE moment problem (N=5 exact; N=9,10 certified); fiber dimensions 0,0,2,10,80,403,1804	Proved / verified	Answers the finite identifiability question (Q1) outright: moment data do not characterize ACUE. New theorem about Tao's object.
Rigidity ladder: 2- and 3-point statistics frozen on the fiber; freedom starts at 4 points	Proved (2-pt) / verified (3-pt)	Localizes every possible AH-distinguishing statistic at order ≥ 4 . New; sharpens Lagarias-Rodgers structurally.
Nyquist conservation law; bound $\frac{1}{2}+2/\pi^2$; hard-core equality face; curve $\Phi(\rho)$	Proved / verified	Exact finite mechanism for where AH multiplicity lives and why limits erase it. Strongest single new theorem in the program.

Transfer theorem: uniform finite gaps = LR non-uniqueness; $\delta^*(N)$ increasing	Proved; frontier open	Turns Q2 into a concrete finite optimization with explicit numbers. A genuine pathway, not yet a resolution.
Quasi-free gap; holonomy TV formula; marked homometry; local blindness	Proved / verified	Exact impossibility perimeter: which statistics can never decide the question. Valuable negative knowledge.
Any statement about actual zeta zeros	None here	No result in this program proves anything about ζ . The 2026 two-thirds theorem is the only zeta theorem in the story, and it is external.

The bottom line. Tao built the ACUE so that the limits of band-limited methods could be studied honestly rather than lamented vaguely. The finite program surveyed here is, to our knowledge, the first sustained mathematical development of that object as an inverse problem: it determines exactly how much the matching moments know (almost nothing: exponential fibers), exactly where they are rigid (all 2- and 3-point statistics), exactly where the AH's characteristic freedom resides (one edge Fourier mode, erased by every scaling limit, tamed only by integrality), and it reduces the field's named open uniqueness question to explicit finite optimization problems with verified numerical frontiers. None of this moves a zeta zero. All of it explains — with theorems instead of folklore — why the unconditional two-thirds theorem is simultaneously a landmark and a wall, and what any attempt to pass the wall must carry: correlation order ≥ 4 , Fourier support beyond the unit band (prime pairs; Siegel-adjacent), or arithmetic positivity that is not a function of correlations at all.

References and pointers

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Verification note: every finite numerical claim quoted in these notes (fiber dimensions, rigidity residuals, the Nyquist identities and constants, the covariance formula, the holonomy total-variation law, the homometric witnesses, $\delta^*(N)$) was independently re-derived or re-computed; methods and worst-case deviations are documented in the companion verification paper. Claims we could not verify are marked conditional above.